- *Global positioning system (GPS) module*: The inclusion of GPS technology revolutionized drone navigation. The GPS module allowed for autonomous waypoint navigation, precise geolocation, and automatic return-to-home (RTH) features. With satellite-based guidance, drones could perform pre-programed missions, follow exact flight paths, and execute precise landings without continuous manual control, enhancing mission reliability and efficiency.
- *Inertial measurement unit (IMU)*: The IMU integrates gyroscopes and accelerometers to monitor the drone's motion, orientation, and angular velocity. This system played a critical role in flight stabilization, compensating for turbulence or sudden changes in wind conditions. By continuously feeding data to the flight controller, the IMU maintained smooth and steady flight even under challenging environmental conditions.
- *Communication and telemetry system*: Second-generation drones featured robust communication and telemetry links for real-time data transmission between the UAV and the ground control station (GCS). These systems used radio frequencies or satellite communication to transmit flight data, video feeds, and sensor readings. The two-way communication link enabled operators to monitor mission progress, adjust flight parameters, and receive live imagery or telemetry feedback from the drone.
- *Power source*: The power system supplied energy to propulsion and electronic subsystems. Depending on the UAV design, drones used fuel tanks, batteries, or hybrid systems. Fuel-based propulsion offered higher endurance for military and long-range missions, while battery-powered systems became popular for shorter, quieter, and